THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 2: **Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solution is prepared by dissolving 30 g of sugar in 120 g of water. The mass percentage is: 1

(a) 25 per cent **(b)** 20 per cent **(c)** 30 per cent **(d)** 12.5 per cent

2. A mixture of two liquids X (b.p. 78 degrees Celsius) and Y (b.p. 118 degrees Celsius) is best separated by: 1

(a) fractional distillation **(b)** sublimation

(c) filtration **(d)** crystallisation

3. A student has a mixture of sand, salt and camphor. The correct sequence of separation is: 1

(a) dissolving, sublimation, evaporation

(b) filtration, sublimation, evaporation

(c) evaporation, sublimation, filtration

(d) sublimation, dissolving and filtering, evaporation

4. How many grams of salt are present in 250 g of a 4 per cent salt solution? 1

(a) 10 g **(b)** 40 g **(c)** 4 g **(d)** 25 g

5. Which technique separates a mixture of iodine and sand? 1

(a) magnet **(b)** decantation **(c)** filtration **(d)** sublimation

6. In fractional distillation the component with the lowest boiling point: 1

(a) stays in the flask **(b)** is collected last

(c) is collected first **(d)** is not collected

7. The volume of ethanol in 250 mL of a 20 per cent (v/v) solution is: 1

(a) 50 mL **(b)** 5 mL **(c)** 20 mL **(d)** 200 mL

8. To separate a mixture of kerosene and water we use: 1

(a) separating funnel **(b)** magnet

(c) distillation **(d)** evaporation

9. Which of the following will pass through a filter paper? 1

(a) sand **(b)** chalk powder

(c) colloidal particles **(d)** iron filings

10. To make 500 g of a 6 per cent solution, the mass of water needed is: 1

(a) 500 g **(b)** 470 g **(c)** 30 g **(d)** 494 g

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the mass of sugar required to prepare 500 g of 8 per cent solution. 2

12. Why is the thermometer bulb placed at the side arm opening in distillation? 2

13. Explain why sea water boils at a higher temperature than pure water. 2

14. A solution is made by dissolving 12 g of salt in 88 g of water. Find the percentage by mass. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A 200 g solution has 36 g salt. Find the percentage. If 100 g water is evaporated find the new percentage. 3

16. A mixture of NaCl and KNO3 is dissolved in hot water and cooled. Which crystallises first and why? 3

17. Why is the temperature of the liquid constant during distillation of a pure liquid and rising during that of a mixture? 3

OR

How will you prepare 250 mL of a 10 per cent (v/v) solution of acetic acid? 3

18. Differentiate between a true solution and a colloid in three points with an example for each. 3

OR

Explain how soap solution and milk differ from copper sulphate solution on shining light. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student has three solutions A, B, C of the same solute with masses of solute 5 g in 95 g water, 10 g in 90 g water and 15 g in 85 g water. Find the percentage of each, rank them and say what happens on mixing A and C (equal masses). 5

OR

Explain, with reasons, how air is separated in industry, and the differences in the boiling points used. 5

20. Describe the separation of a mixture containing iron filings, ammonium chloride, salt and sand in correct order and explain the principle at each stage. 5

OR

Describe an experiment to separate the pigments of a leaf extract by chromatography and state the observations. 5

END OF QUESTION PAPER — Total: 40 Marks